

Conditional Statements Practice Sheet

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This practice sheet contains carefully selected problems to help you strengthen your understanding of conditional statements in Java. To maximize your learning, please attempt each problem on your own before reviewing the solutions. If you don't understand any syntax then, simply search for the syntax on Google and learn about their functionalities.

Problem 1: Check This Number

1.Question:

Write a Java program that reads an integer from the user and prints whether the number is **positive**, **negative**, or **zero**.

Solution:

// **Note:** i've already mentioned in the class that you can use this symbol * instead of Scanner. Again, scan instead of sc. Both cases are considered correct. Choose whatever you prefer.

```
import java.util.*;

public class PositiveNegativeZero {
    public static void main(String[] args) {
        Scanner scan = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = scan.nextInt();

        if (n > 0)
            System.out.println("Number is positive");
        else if (n < 0)
            System.out.println("Number is negative");
        else
            System.out.println("Number is zero");
    }
}
```

Problem 2: FizzBuzz Checker

Question:

Write a Java program that takes a number and prints:

- **FizzBuzz** if divisible by 3 and 5
- **Fizz** if only divisible by 3
- **Buzz** if only divisible by 5
- **Else**, print the number itself

Solutoin:

```
import java.util.Scanner;

public class FizzBuzz {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        if (num % 3 == 0 && num % 5 == 0)
            System.out.println("FizzBuzz");
        else if (num % 3 == 0)
            System.out.println("Fizz");
        else if (num % 5 == 0)
            System.out.println("Buzz");
        else
            System.out.println(num);
    }
}
```

Problem 3: Solve a Quadratic Equation

Question:

Write a program that takes **a**, **b**, and **c** as input and solves the quadratic equation using the discriminant formula.

Solution:

```
import java.util.Scanner;
```

```

public class QuadraticSolver {
    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Solve using  $ax^2 + bx + c = 0$ ");
        System.out.print("Enter value for a: ");
        double a = scanner.nextDouble();

        System.out.print("Enter value for b: ");
        double b = scanner.nextDouble();

        System.out.print("Enter value for c: ");
        double c = scanner.nextDouble();

        // Calculate the discriminant  $D = b^2 - 4ac$ 
        double discriminant = b * b - 4 * a * c;

        System.out.println("\nDiscriminant (D) = " + discriminant);

        // Find the nature of the roots based on the discriminant
        if (discriminant > 0) {
            // Two real and distinct roots
            double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
            double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);
            System.out.println("two real and distinct roots:");
            System.out.println("Root 1 = " + root1);
            System.out.println("Root 2 = " + root2);
        } else if (discriminant == 0) {
            // One real root (repeated)
            double root = -b / (2 * a);
            System.out.println("one real and repeated root:");
            System.out.println("Root = " + root);
        } else {
            // No real roots (complex numbers)
            System.out.println("no real roots.");
        }
    }
}

```

Problem 4: Day of the Week

Question:

Write a program that takes a number (1-7) and prints the corresponding day of the week.

Example: 1 = Monday, ..., 7 = Sunday

Solution:

```
import java.util.Scanner;

public class WeekdayIfElse {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();

        if (day == 1) {
            System.out.println("Monday");
        }
        else if (day == 2) {
            System.out.println("Tuesday");
        }
        else if (day == 3) {
            System.out.println("Wednesday");
        }
        else if (day == 4) {
            System.out.println("Thursday");
        }
        else if (day == 5) {
            System.out.println("Friday");
        }
        else if (day == 6) {
            System.out.println("Saturday");
        }
        else if (day == 7) {
            System.out.println("Sunday");
        }
        else {
            System.out.println("Invalid day");
        }
    }
}
```

```
    }  
  }  
}
```

Problem 5: Grading System

Question:

Write a Java program that takes a score (0-100) and prints a letter grade:

- A: 92 and above
- B: 82 to 91
- C: 72 to 81
- D: 62 to 71
- E: below 62

Solution:

```
import java.util.Scanner;  
public class GradingSystem {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter score: ");  
        double score = sc.nextDouble();  
        char grade;  
  
        if (score >= 92) grade = 'A';  
        else if (score >= 82) grade = 'B';  
        else if (score >= 72) grade = 'C';  
        else if (score >= 62) grade = 'D';  
        else grade = 'E';  
  
        System.out.println("Grade: " + grade);  
    }  
}
```

Problem 6: Find Largest of Three Numbers

Question:

Write a program to take three numbers and print the largest among them.

Solution:

```
import java.util.*;

public class LargestOfThree {
    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter three numbers:");

        System.out.print("First number: ");
        int firstNumber = scanner.nextInt();
        System.out.print("Second number: ");
        int secondNumber = scanner.nextInt();

        System.out.print("Third number: ");
        int thirdNumber = scanner.nextInt();

        // Assume firstNumber is the largest initially
        int largest = firstNumber;

        if (secondNumber > largest) {
            largest = secondNumber;
        }

        if (thirdNumber > largest) {
            largest = thirdNumber;
        }

        System.out.println("The greatest number is: " + largest);
    }
}
```

Problem 7: Leap Year Checker

Question:

Write a Java program to check if a year is a leap year:

- Divisible by 400 → Leap
- Divisible by 100 but not 400 → Not Leap
- Divisible by 4 → Leap
- Else → Not Leap

Solution:

```
import java.util.*;
public class LeapYear {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a year: ");

        int year = sc.nextInt();

        if (year % 400 == 0) {

            System.out.println(year + " is a leap year");

        } else if (year % 100 == 0) {

            System.out.println(year + " is not a leap year");

        } else if (year % 4 == 0) {

            System.out.println(year + " is a leap year");

        } else {

            System.out.println(year + " is not a leap year");

        }

    }

}
```

```
}  
  
}
```

Problem 8: Electricity Bill Calculator

Question:

Calculate the total electricity bill based on units consumed:

- 1-50 units: BDT 0.50/unit
- 51-200: BDT 0.75/unit
- 201-450: BDT 1.20/unit
- 450: BDT 1.50/unit
→ Add 20% surcharge

Soltion

```
import java.util.*;  
  
public class ElectricityBill {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        System.out.print("Enter units consumed: ");  
        int units = scanner.nextInt();  
        double bill = 0;  
  
        if (units <= 50) {  
            bill = units * 0.5;  
        } else if (units <= 200) {  
            bill = 50 * 0.5 + (units - 50) * 0.75;  
        } else if (units <= 450) {  
            bill = 50 * 0.5 + 150 * 0.75 + (units - 200) * 1.20;  
        } else {  
            bill = 50 * 0.5 + 150 * 0.75 + (units - 200) * 1.20 + 20 * bill;  
        }  
    }  
}
```

```

        bill = 50 * 0.5 + 150 * 0.75 + 250 * 1.20 + (units - 450) * 1.50;
    }

    bill += bill * 0.20; // 20% surcharge

    System.out.println("Total Bill: " + bill);
}
}

```

Problem 9: BASIC CALCULATOR [PROJECT]

Question: Create a Basic Calculator from your code-editor

Include the following Features:

- Supports basic operations:
 - Addition
 - Subtraction
 - Multiplication
 - Division
 - Modulus
- Takes two numbers and an operator from the user
- Handles division by zero

Project-Codes:

```

import java.util.Scanner;

public class BasicCalculator {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input: Two numbers
    }
}

```

```
System.out.print("Enter first number: ");

double num1 = sc.nextDouble();
System.out.print("Enter second number: ");

double num2 = sc.nextDouble();

    sc.nextLine(); // Consume leftover newline


// Input: Operator as String
System.out.print("Enter operator (+, -, *, /, %): ");
String op = sc.nextLine();


double result;


// Handling using String and if-else
if (op.equals("+")) {
    result = num1 + num2;
    System.out.println("Result = " + result);
}

else if (op.equals("-")) {
    result = num1 - num2;
    System.out.println("Result = " + result);
}

else if (op.equals("*")) {
    result = num1 * num2;
    System.out.println("Result = " + result);
}

else if (op.equals("/")) {
    if (num2 != 0) {
        result = num1 / num2;
        System.out.println("Result = " + result);
    } else {
```

```

        System.out.println("Cannot divide by zero");
    }
}
else if (op.equals("%")) {
    if (num2 != 0) {
        result = num1 % num2;
        System.out.println("Result = " + result);
    } else {
        System.out.println("Cant perform mod with 0 ");
    }
}
else {
    System.out.println("Invalid operator!");
}
}
}

```

Problem 10: Write The Final Output

Code:

```

public class TracingIfElse {
    public static void main(String[] args) {
        int a = 8, b = 12, c = 5;
        int result = 0;

        if (a > b) {
            result = a + c;
        } else if (b > c) {
            if ((b - a) % 2 == 0) {
                result = b * 2;
            }
        }
    }
}

```

```

        } else {
            result = b + c;
        }
    } else {
        result = c * 3;
    }

    if (result > 20) {
        result = result - a;
    } else if (result == 20) {
        result = result + b;
    } else {
        result = result + c;
    }

    System.out.println("Final Result: " + result);
}
}

```

Solution: *The code will output 16*

{ "So verily, with hardship, there is relief. Verily, with hardship, there is relief." – Surah Ash-Sharh (94:5-6) }